

CodeSync.AI

AI-POWERED COLLABORATIVE DEVELOPMENT PLATFORM

Build faster. Collaborate smarter.

CodeSync.AI is an AI-powered collaborative development platform that helps developers generate code, manage project files, preview applications, and work together through controlled project access.

AI Code Generation	Generate code from natural-language prompts.
Project Workspace	Manage files and preview project output.
Team Collaboration	Share projects with different access levels.
Role-Based Access	Owner, Editor and Viewer permissions.

Right role. Right permissions.

👑 Owner

Manages project, its members & sharing permissions, with full project access.

📝 Editor

Works with project files, development tools according to permissions granted.

👁️ Viewer

Views available project content without modifying files & project members.

One platform. Infinite reach.

🗨️ Streaming AI Chat	Receive AI responses as they are generated.
📁 File Management	Explore project files and inspect generated code.
🔍 Live Preview	Preview supported web project output in workspace.

Behind the intelligence. Built for collaboration.

React	Interactive user interface
Spring Boot	Backend services and business logic
PostgreSQL	Persistent data storage
AI	AI-powered code generation
JWT & Spring Security	Authentication and access control
Server-Sent Events (SSE)	Streaming AI responses

From prompt. To working code.

Create	Create a project or open a shared workspace.
Prompt	Describe the required functionality to the AI.
Generate	Receive AI-generated code through streaming responses.
Manage	Explore files and review generated changes.
Preview	View the supported application output.
Collaborate	Share the project with role-based access.

Secure by design. Controlled by roles.

🔑 JWT-based authentication

👥 Role-based project permissions

🔒 Protected backend APIs

🛠️ Secure configuration using environment variables

Challenge faced. Solution engineered.

Debugging and Development Time

Challenge: Resolving repeated bugs and managing dependencies between modules increased development time.

Solution: I used systematic debugging, tested modules individually, and fixed issues step by step until the application was completed.

Learning: Structured debugging and persistence are essential for reliable software development.

Challenges faced. Solutions engineered.

AI Response Streaming

Challenge: Displaying AI responses in real time without waiting for the complete response

Solution: I implemented Server-Sent Events (SSE) to stream AI responses progressively to the frontend.

Learning: Efficient communication mechanisms improve application responsiveness.

Challenge faced. Solution engineered.

Managing a Large Project

Challenge: Managing backend logic and frontend integration in a large full-stack application was challenging.

Solution: I documented implementation steps, broke the project into smaller modules, and followed a structured development workflow.

Learning: Modular development and technical documentation improve maintainability.

Built to evolve. Ready for more.

Google Authentication

Integrate Google OAuth 2.0 to enable secure and easy user sign-up and login.

Access Request System

Allow Editors, Viewers to request additional permissions from the project owner. The owner can approve or reject requests, similar to social media follow request system.

Enhanced Real-Time Collaboration

Enable multiple users to collaborate on the same project with real-time updates.

Built to evolve. Ready for more.

AI Code Validation and Testing

Validate AI-generated code and identify errors through automated testing.

Extended Technology Support

Support additional programming languages and frameworks to expand development possibilities.

Integrated Code Compiler & Execution

Integrate a code compiler and execution environment to compile and run code directly within the application, helping users identify errors and verify outputs without switching to external tools.